

Task 1 — Power System Design

Scenario: a 3S Li-ion battery (12.6V full charge) powers Load A — a Songle SRD relay coil — and Load B — an HC-SR04 ultrasonic sensor. Source datasheets used below: Songle SRD relay (coil data), ElecFreaks HC-SR04, Texas Instruments LM1117/TLV1117 (LDO), and Texas Instruments LM2596 (SIMPLE SWITCHER buck converter).

Part 1 — Identify the Needs

1.1 Voltage and current required by each load

Load	Voltage	Current
Load A — Relay coil (SRD-12VDC, High Sensitivity)	12V DC	30mA
Load B — HC-SR04 Ultrasonic Sensor	5V DC	15mA

1.2 Is the battery voltage higher or lower than what's needed?

The 3S Li-ion battery supplies 12.6V at full charge. Compared against each load:

Load	Needs	Battery vs. Need
Relay coil	12V	Higher, by 0.6V only
HC-SR04	5V	Higher, by 7.6V

In both cases the battery voltage is higher than what the load needs, so both outputs require step-down regulation — but the size of the step-down is very different for each load, which drives different regulator choices in Part 2.

1.3 Do we need reverse-polarity protection?

Yes. If the battery is connected backwards, the reverse voltage can permanently damage the regulator ICs (most are not rated to survive a reversed input) as well as the relay coil and the HC-SR04 module if it reaches them directly. A simple protection component at the battery input is inexpensive compared to the cost of replacing damaged parts.

Part 2 — Choose the Components

2.1 Regulator type for each output

Load A — Relay (12V): the gap between the battery (12.6V) and the target (12V) is only 0.6V. An LDO is the better fit here: the power wasted as heat is small ($\Delta V \times I \approx 0.6V \times 0.03A \approx 18mW$), and an LDO is simpler, cheaper, and quieter than a buck converter — adding switching-converter complexity for such a small drop isn't justified.

Load B — HC-SR04 (5V): the gap here is large ($12.6V - 5V = 7.6V$). A Buck converter is required: an LDO at this drop would waste $7.6V \times 0.015A \approx 114mW$ and run at roughly 40% efficiency, while a buck converter can reach 85–95% efficiency for the same job.

2.2 Protection circuit

A single Schottky diode in series with the battery's positive line, placed before the branch to both regulators, protects both loads with one component. Schottky is chosen over a standard diode because of its lower forward-voltage drop ($\sim 0.3V$ vs $\sim 0.7V$), which matters most on the relay's already-tight 0.6V headroom.

Part 3 — Check the Datasheet

3.1 Load A — LDO for the relay coil (12V)

Parameter (800 mA)	LM1117	TLV1117	Unit
Input voltage range (max)	15	15	V
Load regulation accuracy	1.6	1.6	%
PSRR (120 Hz)	75	75	dB
Recommended operating temperature	0 – 125	-40 – 125	°C

✓ LM1117 fits this load

Input voltage range (max): 15V — comfortably covers the 12.6V battery. Output is adjustable via the ADJ/GND pin with a resistor divider, so it can be set to 12V. Current rating: 800mA, far above the relay's 30mA.

Caveat: LM1117-family LDOs typically need roughly 1–1.2V of headroom (dropout) between input and output to stay in regulation — this datasheet excerpt does not state the dropout voltage. With only 0.6V of margin between the battery and 12V, this should be checked against the dropout spec before finalizing the part; if the dropout exceeds 0.6V, the battery would need to stay nearer full charge for the LDO to regulate correctly, or a lower-dropout part should be used.

3.2 Load B — Buck converter for the HC-SR04 (5V)

The TPS40222 checked previously was rejected because its recommended input (4.5–8.0V, 10V absolute max on AVIN/PVIN) does not cover the 3S battery's 9–12.6V range. The LM2596 SIMPLE SWITCHER 3-A step-down regulator was selected instead and checked against the same requirements:

Requirement	Needed	LM2596 Datasheet	Result
Input voltage range	9 – 12.6 V	Up to 40V	Covers battery
Output voltage	5 V (fixed target)	Fixed 3.3/5/12V or adj. 1.2–37V $\pm 4\%$	LM2596-5.0 gives 5V directly
Output current	≈ 15 mA	Rated 3A output load current	Very large margin ($\approx 200\times$)
Switching / efficiency	n/a	150 kHz fixed oscillator, high efficiency	Suits small, simple design

✓ LM2596 fits this load

Input voltage range up to 40V comfortably covers the full 9–12.6V range of the 3S battery across its whole discharge curve, unlike the rejected TPS40222. The fixed LM2596-5.0 version outputs 5V directly with no resistor divider needed, and its 3A current rating leaves enormous headroom above the HC-SR04's 15mA draw. The datasheet's own typical application circuit uses a 12V unregulated DC input driving a 5.0V/3A output — essentially the same configuration as this design.

Caveat: this excerpt doesn't list the part's minimum input/dropout voltage explicitly; the full datasheet lists a dropout of roughly 1–1.2V at rated current. Even at the battery's lowest expected voltage (≈ 9 V, 3S near cutoff) this leaves a wide margin above the 5V output, so it should not be a limiting factor, but it is worth confirming against the full electrical characteristics table before finalizing.

Summary of Part 3: the LDO choice for the relay (LM1117-family) checks out on voltage and current, with one dropout caveat to confirm. The buck converter choice for the HC-SR04 also checks out — replacing the TPS40222 with the LM2596 resolves the input-voltage mismatch found earlier, since the LM2596's 40V input rating and 3A output easily cover the 3S battery and the HC-SR04's small current draw.

Part 4 — Simulate in Proteus

Both regulator circuits were built in Proteus (Schematic Capture / ISIS) using the same component values shown in the datasheets' typical application circuits, then simulated together with a shared 12.6V source (BAT1/BAT2, two 12.6V cells in series representing the 3S pack) to confirm both outputs regulate correctly and stay stable.

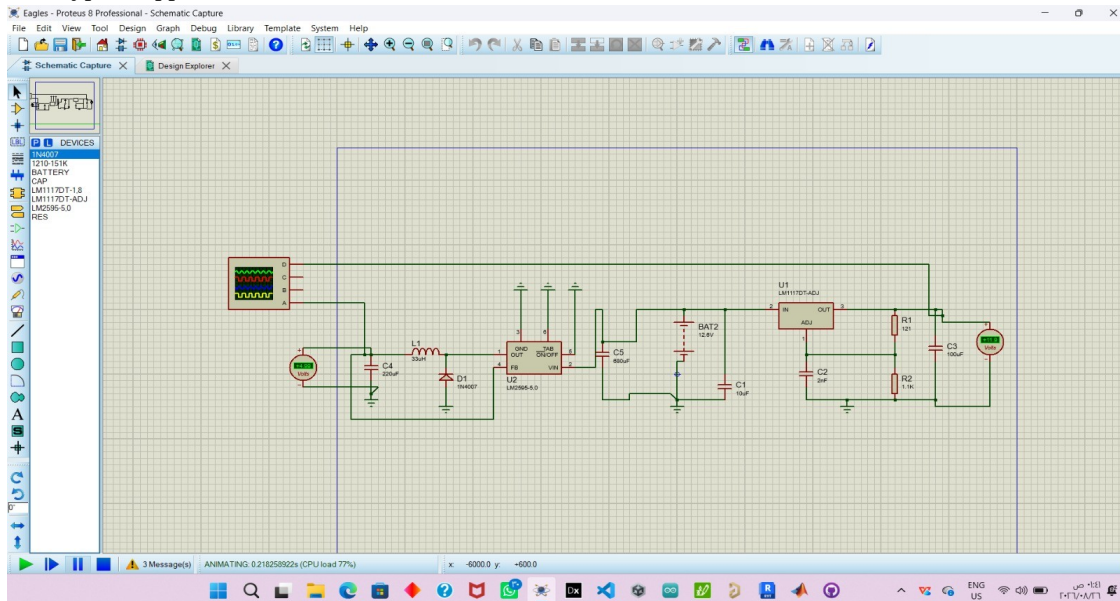
4.1 Circuit build

- Load B branch (U2 — LM2595-5.0): VIN from the battery rail, C4 = 220 μ F input capacitor, L1 = 33 μ H inductor, D1 = catch diode (1N4007 used as a substitute for the 1N5824 Schottky, since the exact Schottky part was unavailable in the local library), C5 = 680 μ F output capacitor. ON/OFF (pin 5) and TAB (pin 6) tied to GND as required.
- Load A branch (U1 — LM1117DT-ADJ): IN from the battery rail, C1 = 10 μ F input capacitor, C2 = 2nF and R1 = 121 Ω / R2 = 1.1k Ω forming the ADJ feedback divider to set the output near 12V, C3 = 100 μ F output capacitor.
- A DC voltmeter was placed on each regulator's output node to read the settled voltage, and a 4-channel oscilloscope was wired into the schematic for waveform capture.

4.2 Simulation result

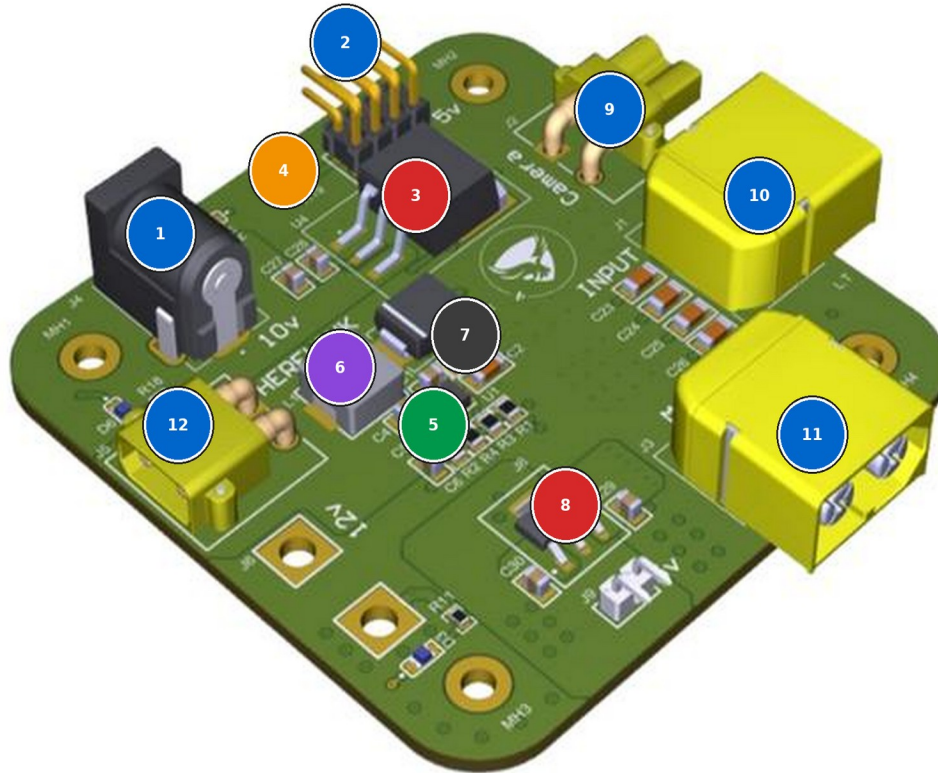
Branch	Target	Simulated (settled)	Result
Load B — LM2595-5.0	5.0 V	4.98 V	Within ~0.4% — stable
Load A — LM1117DT-ADJ	12.0 V	11.9 V	Within ~0.8% — stable

Both readings held steady while the simulation ran (no drift or oscillation on the DC voltmeters), confirming the regulator + capacitor/inductor selections from Part 3 produce a stable output under simulation, matching the datasheets' typical application circuits.



Task 2 — PCB Component Identification

The numbered markers on the board below identify one representative example of each component category requested in the task — capacitors, resistors, inductors, voltage regulators, connectors, and diodes. Each number is explained in the table underneath, colour-coded by category.



Component Legend

#	Component	Function	Why It's Needed
1	DC Barrel Jack (12V / 10V input)	Main external power-input connector; feeds the raw battery/adaptor voltage onto the board.	Gives a mechanically secure, polarized way to bring external power in without a soldered wire joint.
2	5-Pin Header	Board-to-board interface exposing the regulated 5V rail and control/data lines.	Lets external modules connect to the board's power and signals without a dedicated cable harness.
3	Main Regulator IC	Switching (buck) regulator — steps the input voltage down to the regulated 5V rail (per the nearby silkscreen).	The input supply is higher than the board's logic needs; this IC does the step-down conversion efficiently instead of wasting the difference as heat.
4	Ceramic Capacitors (cluster)	Input/output decoupling — filters high-frequency noise and smooths the supply rail next to the regulator.	A switching regulator needs a clean, low-impedance rail at its pins; without these caps, ripple and noise would degrade the output.
5	Resistor Array	Sets bias points and feedback ratios for the nearby regulator/logic (voltage-divider and current-limiting roles).	Fixes the exact output voltage the regulator holds and protects signal pins from excess current.

#	Component	Function	Why It's Needed
6	Power Inductor (shielded)	Energy-storage element of the switching regulator — stores energy while the internal switch is on, releases it to the output while it is off.	This is what makes step-down conversion efficient; without it, the voltage difference would have to be dropped resistively and wasted as heat.
7	Diode	Rectifies or protects current flow at this node (e.g., blocks reverse current or provides a freewheeling path for the inductor).	Protects the regulation stage from damage on reversed input and/or completes the switching-converter current path.
8	Secondary Regulator IC	Second regulation stage that further conditions the rail to a clean, logic-level output (5V) for the output connector.	Isolates sensitive downstream loads from switching noise generated by the first-stage regulator and guarantees a stable output rail.
9	Camera Connector	Dedicated interface (power + data) for an attached camera module.	Provides a standardized, keyed connection so a camera module can be added without custom wiring.
10	XT Connector (large, board-mounted)	Secondary power-input/output connector — a bullet-style two-pin XT connector for a high-current external supply or battery pack, separate from the barrel-jack input.	The bullet-style polarized pins give a low-resistance, secure power connection for high-current loads without a soldered wire joint.
11	XT Connector (small)	Secondary power tap connector — a smaller bullet-style two-pin XT connector delivering the board's regulated output rail to an external peripheral.	The bullet-style polarized pins give a secure, low-resistance power connection for lighter-current peripherals without a soldered wire joint.
12	Small Barrel/JST Connector	Secondary power tap near the 12V silkscreen label — likely feeds an auxiliary load directly from the input rail.	Lets one additional peripheral draw power without adding a second regulator stage.

Reading the board as a whole: power enters through the barrel jack (1), is stepped down by the main regulator (3) with its supporting inductor, capacitors, and resistors (4, 5, 6), protected/rectified by the diode (7), optionally conditioned again by a second regulator (8), and finally exposed to peripherals through the header, camera, and XT connectors (2, 9, 10, 11, 12).

Task 3 (Bonus) — Multi-Sensor Schematic Design

Design Requirements

- MCU: Arduino Nano (5V logic)
- Sensors: two MPU6050 IMUs sharing one I2C bus
- Storage: SD card module (SPI, 3.3V logic only)
- Wireless: nRF24L01+ module (SPI, 3.3V logic only)
- Power source: single 3S LiPo battery (11.1V–12.6V)

Three constraints had to be satisfied: (1) both MPU6050s must be individually readable despite sharing one I2C bus with identical default addresses; (2) the SD card and the nRF24L01+ share the SPI bus with the Nano without interfering with each other; (3) power must be correctly derived from the 3S LiPo for every component.

Design Decisions

1. Power tree (3S LiPo → 5V → 3.3V)

The 12.6V pack is first stepped down by a buck converter (U4, LM2596) to a regulated 5V rail that feeds the Arduino Nano directly. That same 5V rail then feeds an LDO (U6) which produces the 3.3V rail required by both MPU6050 modules, the SD card holder, and the nRF24L01+ — none of these can tolerate 5V on their logic/supply pins.

2. I2C address conflict (two identical MPU6050s)

Each MPU6050 has one address pin, AD0, which selects between two default I2C addresses. U5's AD0 is tied to GND, giving it address 0x68; U2's AD0 is tied to 3.3V, giving it address 0x69. Both sensors then share the same SDA/SCL lines and can be addressed independently in software. A single pair of 4.7kΩ pull-up resistors from SDA and SCL to 3.3V is required for the bus to operate reliably.

3. Shared SPI bus (SD card + nRF24L01+)

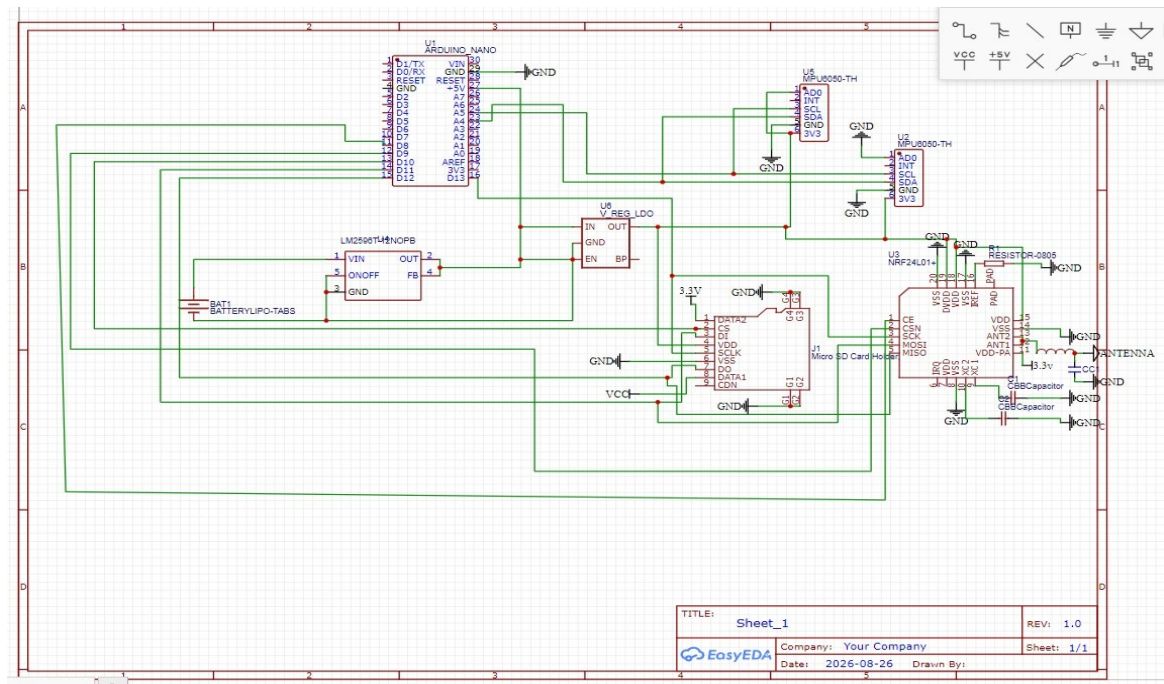
MOSI, MISO, and SCK are common to both peripherals and the Nano. Each peripheral gets its own dedicated Chip Select line so the Nano can address one at a time: D10 → SD card CS, D9 → nRF24L01+ CSN. The nRF24L01+ additionally needs a CE line (D8) to switch between standby and active RX/TX mode.

4. nRF24L01+ support circuitry

The schematic uses the raw 20-pin nRF24L01+ die symbol rather than an 8-pin breakout, so several supporting elements are required beyond the SPI/CE/CSN lines that a ready-made module would normally hide:

- All VDD/DVDD pins tied to the 3.3V rail; all VSS pins and the thermal PAD tied to GND.
- A 16MHz crystal across XC1/XC2, each side additionally decoupled to GND through a small load capacitor (C1, C2).
- IREF biased to GND through a precision resistor (R1) as specified by the Nordic datasheet.
- An antenna matching network (inductor + capacitor) between ANT1/ANT2 and the antenna feed point, so the 2.4GHz output is presented with the correct load impedance. Final component values depend on the antenna type used and were left as a reference network per Nordic's application note rather than a fully tuned design, which is outside the scope of this schematic-level task.

Final Schematic



Connection Tables

Power

From	To	Note
BAT1 (+)	U4 VIN (pin 1)	Raw 3S LiPo, 11.1–12.6V
BAT1 (-)	GND (common)	+ U4 GND (pin 3)
U4 ONOFF (pin 5)	GND	Enables the LM2596 (active-low)
U4 OUT (pin 2)	Nano +5V (pin 27)	Regulated 5V rail
U4 FB (pin 4)	U4 OUT (pin 2)	Feedback tied to output (fixed 5V version)
U4 OUT (5V rail)	U6 IN	LDO input
U6 EN	U4 OUT / IN (5V rail)	Enables the LDO
U6 GND	GND (common)	
U6 OUT (3.3V rail)	U2, U5, J1 VDD, U3 VDD/DVDD	Common 3.3V distribution

I2C bus (MPU6050 ×2)

From (Nano)	To
A4 (SDA)	U2 SDA + U5 SDA (shared)
A5 (SCL)	U2 SCL + U5 SCL (shared)
—	4.7kΩ pull-ups from SDA and SCL to 3.3V
U5 AD0	GND → address 0x68
U2 AD0	3.3V → address 0x69

SPI bus (SD card + nRF24L01+)

From (Nano)	To
D13 (SCK)	J1 SCLK + U3 SCK

